12TH CHEMISTRY · CHAPTER 20 · CHAPTER TEST

Aromatic Hydrocarbons

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 20	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. The C–C bond length in benzene is:

(a) 120 pm (b) 133 pm
(c) 154 pm (d) 139 pm

2. The resonance energy of benzene is about:

(a) 120 kJ/mol (b) 152 kJ/mol
(c) 208 kJ/mol (d) 360 kJ/mol

3. The electrophile in the nitration of benzene is the:

(a) Nitronium ion (b) Nitrate ion
(c) Nitrite ion (d) Hydrogen sulphate ion

4. Which group is meta directing?

(a) Methyl (b) Hydroxyl
(c) Nitro (d) Alkoxy

5. Side-chain oxidation of toluene gives:

(a) Phenol (b) Benzoic acid
(c) Benzaldehyde (d) Cyclohexane

6. Complete hydrogenation of benzene gives:

(a) Cyclohexene (b) Hexane
(c) Cyclohexane (d) Toluene

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Define resonance energy of benzene.
ii. Why are all C–C bonds in benzene of equal length?
iii. What is a sigma complex?
iv. Why does benzene prefer substitution to addition?
v. What is the role of aluminium chloride in Friedel-Crafts reactions?
vi. What is a benzylic hydrogen?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Explain the atomic orbital structure of benzene.
(b) Describe the mechanism of nitration of benzene.

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